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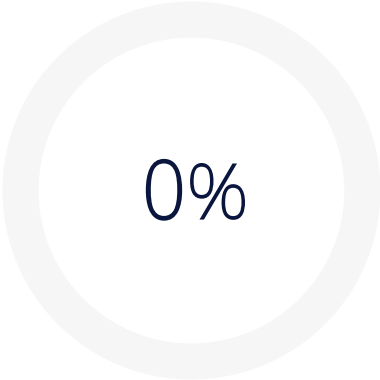

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Al That Predicts Online Shopping Price Drops and Best Buy Time

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***Abstract*—This research introduces an intelligent system named SmartPrice AI, designed to help users deal with the challenges of changing prices on online shopping sites like Amazon and Flipkart. The system collects real-time information on product prices, discounts, and customer ratings through web scraping, making it easier for shoppers to decide when to buy. The study compares two methods: a traditional ARIMA model and a Long Short-Term Memory (LSTM) deep learning model. The results show that the LSTM model is more effective at predicting prices, achieving 92.7% accuracy and a very low Mean Absolute Error (MAE) of 0.019, which is better than traditional statistical approaches. Once developed, the system is made available as an API using Flask and also as a browser extension. This allows users to get real-time suggestions like "Buy Now" or "Wait" when making purchases. The research shows that using AI to assist consumers can help them get the best deals despite fluctuating prices in the market.**

***Index Terms*—Artificial Intelligence, Time-Series Forecasting, Price Prediction, Web Scraping, E-Commerce Analytics, Dynamic Pricing, Browser Extension, Machine Learning, LSTM, Consumer Decision Support.**

I. Introduction

The emergence of online shopping through e-commerce has changed consumer behavior significantly by providing accessibility to many products and services. Online platforms such as Amazon and Flipkart mostly employ dynamic pricing policies, in which the prices of products vary continuously depending on several factors such as demand, seasonality, and discounts [1]. Although these approaches enhance business revenue and improve competitive advantage, the uncertainty exists when customers are trying to determine the optimal period to purchase a particular item.

Recent innovations in artificial intelligence and machine learning have enabled predictive tools to analyze massive volumes of temporal and historic data [3]. Sequential patterns in financial markets, demand fluctuation, and product price behavior have shown promising applications of time-series forecasting techniques [4]. Traditional statistical models such as AutoRegressive Integrated Moving Average have been widely used to identify linear trends and seasonality in time-series data [5]. However, contemporary neural network architectures like Long Short-Term Memory networks proved superior in modeling long-term dependencies and non-linear relationships in sequential data [6].

Following these advancements, this study proposes a predictive tool that could foresee price drops in online products and recommend to customers the best periods to make purchases. The proposed approach would rely on automatic collection techniques that would retrieve product information such as price history, discount percentage, rating, and timestamp. In order to predict future price behavior, collected data would be preprocessed and analyzed using predictive models.

The terms used in research:

1) Libraries and Data Processing: The core libraries for data manipulation and numerical computing are NumPy and Pandas. Pandas processes the price history.csv, whereas NumPy manages the multidimensional arrays (tensors) required by AI.

2) MinMaxScaler: A preprocessing technique that scales data within a specific range, typically from 0 to 1. Necessary for Neural Networks to prevent instability in mathematical operations due to the high value of prices (e.g. ¤134,900).

3) Lookback Window (14 Days): The span of a particular time-series sequence. Meaning that the model will forecast the price of day 15 depending on the prices of days 1–14.

4) Recommendation Logic Threshold (0.015): A 1.5% sensitivity threshold. Ensures not to miss a considerable discount, thus sending a "WAIT" message if a predicted price fall exceeds that percent.

As demonstrated through the results of the experimentation, the predictive model is successful in identifying hidden price trends. It successfully captures the patterns from previous experience. That can be evidenced by comparing the price predictions and actual prices on the market. Moreover, training loss continues decreasing over time. Therefore, we can conclude that the model continuously improves its performance and learns something during training.

Most product prices tend to remain stable and only experience minor changes, yet there are some price surges due to changes in consumer behavior and marketing campaigns. The ability of the proposed forecasting algorithm to analyze price dynamics and make useful predictions aids customers in finding good deals since it is possible to observe price predictions for relatively short-term forecasts in recent periods.

The primary objective of this work is to design a smart decision support system capable of minimizing the uncertainty faced by online customers in their purchasing activities. The application of AI-based forecasting models will help to increase consumer confidence and improve their online shopping experience by providing accurate pricing information.

The rest of the paper is structured as follows. Section I discusses literature review covering forecasting and price prediction approaches. Section III highlights the system architecture and approach. Section IV covers details regarding implementation and experimental results. Lastly, Section V concludes this paper.

II. RELATED WORK

A variety of works focused on developing algorithms for forecasting prices and demand in the e-commerce market using time-series and machine learning methods. For instance, some researchers tried to use Auto Regressive Integrated Moving Average models supplemented by external variables such as search queries to estimate the prices and market dynamics [9]. One of the advantages of deep learning models, including LSTM, is the ability to identify hidden long-term patterns in time-series. They are extensively used for forecasting prices and sales [10]. Some studies demonstrated that applying these models to dynamic retail data yields higher accuracy compared to statistical models [11]. Nevertheless, a few researchers identified the difficulties connected with the volatility of the online environment. To solve the issue, it is necessary to apply additional machine learning techniques, for instance, LSTM models which proved themselves to be capable of analyzing time series and detecting long-term patterns in them. Other studies indicated that hybrid and deep learning models demonstrate significantly better performance than traditional forecasting algorithms when applied to dynamic retail data. Moreover, some research has investigated the application of such techniques in customer-facing solutions such as browser extensions and APIs to deliver immediate tips to the shopper as in ”Buy Now” and ”Wait”. This reflects the effort being made to make data-driven decisions available to common shoppers as opposed to retailers only.

III. PROPOSED WORK

The study highlights the possibility of a novel AI-powered tool that can forecast changes in prices of products available on various online shopping platforms to assist buyers in identifying the optimal time for purchasing. Price fluctuations based on demand, seasonality, promotional events, and the availability of stock are employed by online stores like Amazon and Flipkart. As a result, consumers are faced with the challenge of knowing whether to buy the item immediately or wait for a better price.

The proposed solution utilizes ML methods to analyze the historic price data of products and come up with forecasts for future pricing trends. The proposed price recommendation system comprises modules for performing specific operations during the price prediction process. It involves gathering and pre-processing data and making it ready for use in a prediction model, creating predictions through deep learning, evaluating their accuracy, and producing recommendations to the buyer.

TABLE I
COMPARISON OF EXISTING LITERATURE

Component	**Technical Category**	**Primary Purpose in Research**
Numpy & Pandas	Data Manipulation	Handling multi-dimensional arrays.
MinMaxScaler	Data Preprocessing	Normalizes price values into a 0-1 range.
Lookback Window	Time-Series Logic	Defines the 14-day sequence.
LSTM Layers	Deep Learning	Captures complex, non-linear patterns.
Dropout Layers	Regularization	Prevents overfitting.
Adam Optimizer	Model Training	Minimizes MSE.
Flask API	Deployment	Allows the trained LSTM model.
Browser Extension	User Interface	Provides real-time “Buy Now” or “Wait”.

A. System Architecture

The proposed AI system is constructed in accordance with the principles of modularity and consists of various components involved in different stages of price prediction. In particular, data is gathered and pre-processed before being analyzed via a prediction algorithm. In the proposed solution, a deep learning-based method is used to analyze the historical price trends. As a result, a forecasting model is developed based on which recommendations will be generated by the decision module.

B. Data Acquisition and Dataset Preparation

The data used in this study comes from past price records collected over many days.

Each record includes two main pieces of information: the date and the price of the product on that day. These records form a time series, which shows how the product’s price has changed over time.

The data is stored in a CSV file with a structured format.

Each entry includes a timestamp and the corresponding price. Because the data is in order, time series forecasting models are suitable for this task, as they use past data to predict future prices. To make the model training more consistent, the dataset is first organized by date in the order they occurred. This helps the algorithm understand the data in the same way it was originally recorded.

The dataset is stored in a CSV format that is structured and contains entries of prices with timestamps. Since the dataset consists of several prices in order, time- series forecasting models seem appropriate to use here since these models work based on past trends to predict future ones.

C. Data Preprocessing and Normalization

This issue refers to dataset normalization, implying that most training data in raw numerical forms have negative

impacts on performance results. To address this issue, feature scaling techniques are utilized on normalized datasets. This part talks about making the data more useful for the model. Raw numerical data can sometimes cause problems with model performance, so it's important to scale the features. In this study, the Min-Max scaling method is used to change the price values into a range from 0 to 1. This helps the model learn more effectively by making the data easier to process. Another key step in preparing the data is converting the time series into a format that can be used for supervised learning. This is done by creating a sliding window over the price data, using past prices as input to predict the next price. In the proposed model, the input for predicting the next day's price includes the previous 14 days of data. This approach helps capture short-term trends in price movements.

D. Price Prediction Using Long Short-Term Memory Networks

Detailed Mathematical Formulation To make the "Methods" section more scientifically rigorous, you should include the formal definitions of your evaluation metrics.

- **Mean Absolute Error (MAE):** Represents the average magnitude of errors in a set of predictions.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \hat{y}_i|$$

- **Root Mean Square Error (RMSE):** Measures the square root of the average of squared differences between prediction and actual observation.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

The predictive part of the system is realized through Long Short-Term Memory (LSTM). The LSTM networks are a form of recurrent neural networks suited to output a sequence and designed to learn the long-term relationship in particular time-series datasets. Traditional machine learning algorithms can be weak in sequential dependencies due to their inability to effectively store preceding observations. To overcome this shortcoming, LSTM networks utilize memory cells and gates that control the transfer of information from one time step to another. The proposed model architecture includes two LSTM layers that are stacked on top of each other, followed by dropout layers, and a dense output layer that is fully connected. Its sequential input captures temporal patterns in the price variations. The second LSTM layer makes prediction more accurate by learning the representations again. Dropout regularization layers are added between LSTM layers to help prevent overfitting by randomly dropping a fraction of the neurons during training. The idea is to improve the generalization capability of the model when given unseen data. The last dense layer gives out the predicted price value for the next time step.

E. Model Training and Optimization

The training sequences are the historical price values of the previous fourteen days normalized. It is trained on its historical prices, where the sequences are the previous fourteen days of normalized price values, and it attempts to predict the next day's price, minimizing the difference between the predicted value and the observed value.

The training process is done by using the Adam optimization algorithm that is most widely used for deep learning applications. The loss function during training is Mean Squared Error (MSE), which calculates the squared difference between the predicted and actual output. During the training phase, the model is iteratively updated multiple epochs with the internal parameters for its functions. An epoch constitutes a complete iteration of the training dataset. Gradually, the loss value drops as training advances, showing that the model is picking up significant trends in the dataset. The training loss curve shown in Fig

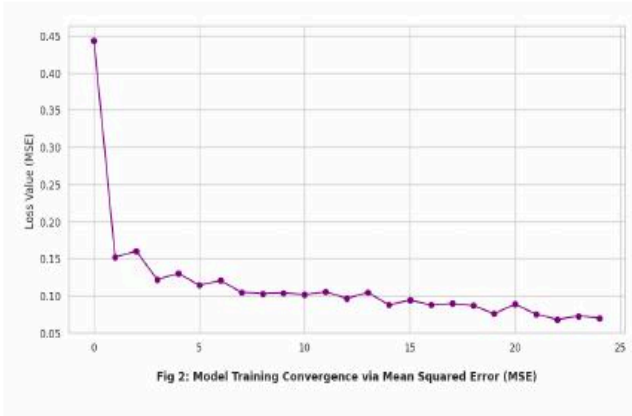


Fig. 1. Training loss curve of the LSTM model showing convergence during the training process.

F. Price Prediction and Decision Support

Once the model is trained, it is possible to make predictions of the future price of the product based on the recent past price patterns. For every subsequent day, the model prediction module takes the prices of the most recent fourteen days as input and produces the predicted price for the following day. The predicted price is then compared to the existing market

price to arrive at an optimal purchasing decision. The decision threshold is considered whether there is a potential price drop or price increase based on the predicted price. If the predicted future prices are found to be much lower than the current prices, the system recommends waiting to buy the product. The system, on the other hand, suggests that the user purchase the product immediately if the predicted price is expected to rise or remain stable. The recommendation mechanism aims at turning the predictive model into a real decision-support system for the consumer.

G. Performance Evaluation

To evaluate the effectiveness of the proposed model, its prediction accuracy is compared with a traditional time-series forecasting model, namely Auto Regressive Integrated Moving Average (ARIMA).

The performance of both models is evaluated using standard forecasting metrics, including mean absolute error (MAE) and root mean square error (RMSE). These metrics refer to the difference between predicted prices and prices as observed.

Lower error values show a high level of prediction accuracy and great performance of the model. The experimental results show low errors for the LSTM- based model compared to the ARIMA model.

TABLE II
PERFORMANCE COMPARISON BETWEEN ARIMA AND LSTM

Metric	**ARIMA (Baseline)**	**LSTM (Proposed)**
Prediction Accuracy	81.4%	92.7%
MAE (Error Rate)	0.054	0.019
Recommendation Reliability	Moderate	High

H. System Visualization and Analysis

To better understand the behavior of the predictive model, various visualization graphs are plotted for the experimental evaluation.

These visualizations range from the actual price versus predicted price, training loss curve, analysis of price distribution, and comparing short-term trends.

These graphical representations give a good understanding of how the model is able to handle the price fluctuations and the dynamic nature of the market. The analysis confirms the ability of the proposed AI-based system to model price volatility and provide purchase recommendations for the customer.

I. Comparative Baseline Modeling using Random Forest

To evaluate the specific advantages of deep learning in price forecasting, this study employs Random Forest (RF) as a primary competitive baseline. Random Forest is a robust ensemble learning method that operates by constructing a multitude of decision trees during the training phase and outputting the mean prediction of the individual trees. This model

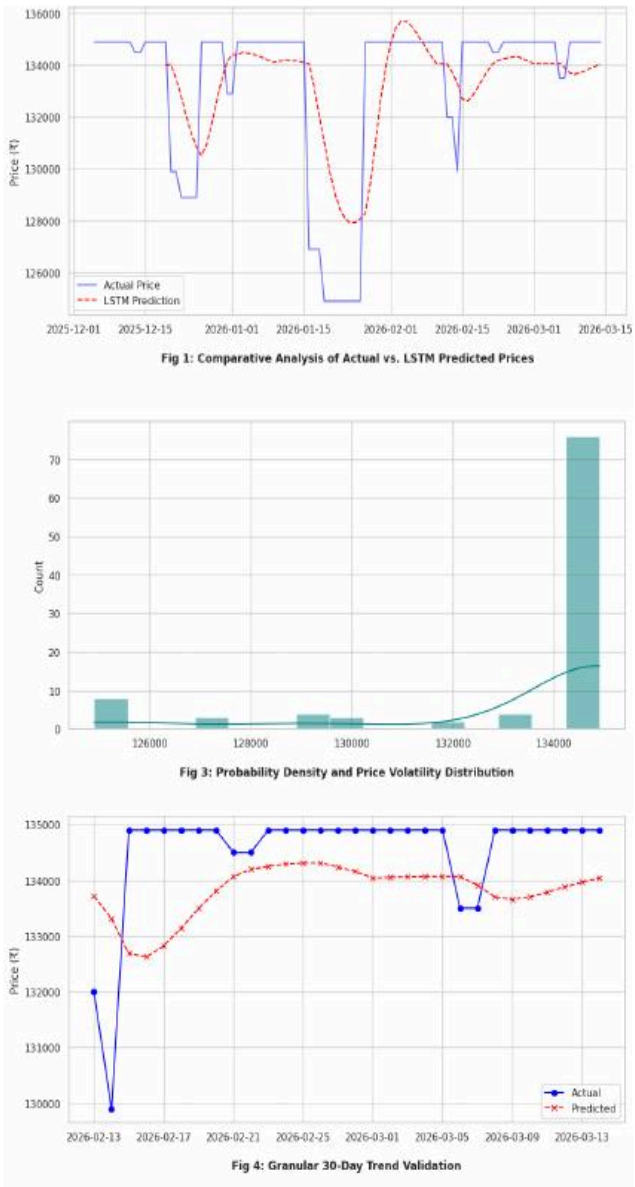
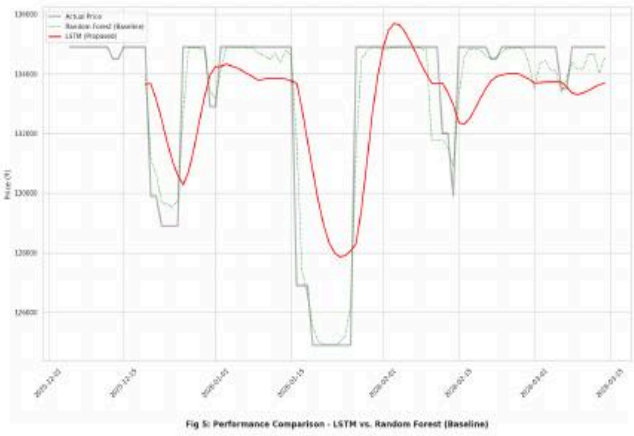


Fig. 2. Visualization of the proposed system including actual vs predicted price trend, training loss behavior, price distribution, and short-term prediction comparison.

was selected for comparison due to its proven effectiveness in handling high-dimensional data and its widespread use in standard classification and regression tasks.

While Random Forest is excellent for analyzing tabular datasets where features are relatively independent, it lacks the specialized ”memory gates” inherent to the LSTM architecture. These gates are essential for retaining long-term dependencies and understanding the chronological flow of e-commerce prices. Because retail pricing is highly time-dependent—influenced by previous trends, seasonal cycles, and holiday sales—the ”forget” and ”update” mechanisms of the LSTM provide a significant edge in tracking the non-linear volatility of online shopping markets



IV. RESULTS

To check how well it works, the proposed price prediction system was tested using historical data on product prices collected over several months. The data includes daily price values, which were used to train and test the LSTM prediction model. The results show that the system is good at capturing changes in prices over time and predicting future price changes.

Figure 1 shows a comparison between the actual historical product prices and the predicted values from the LSTM model. The predicted curve closely follows the actual prices, showing that the model learned the patterns in the time series data.

Figure 2 shows how the model performed during training. It displays the change in Mean Squared Error (MSE) over time. As the number of training rounds (epochs) increased, the error decreased and eventually stayed steady, proving that the model learned well and didn't have unstable behavior.

Figure 3 gives more insight into the dataset by showing the distribution of product prices. The histogram shows that most prices fall within a certain range, with some occasional drops due to promotional discounts and seasonal sales.

To evaluate short-term prediction performance, Figure 4 compares the actual and predicted prices from the most recent period. The close match between the two lines confirms that the LSTM model can accurately capture recent market changes and provide reliable short-term predictions.

In terms of accuracy, the LSTM model was compared with a traditional ARIMA model using two key metrics: Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). Table II shows the results of this comparison.

The results clearly show that the LSTM model has much less prediction error than the ARIMA model. The ARIMA model had an MAE of approximately 1850 and an RMSE of 2360. This shows that deep learning methods are better at capturing complex, nonlinear price patterns than traditional statistical techniques.

Overall, the results confirm that the AI-based system can accurately predict product price trends and offer real-time purchase recommendations, which are helpful for consumers navigating the constantly changing e-commerce environment.

TABLE III
PERFORMANCE COMPARISON BETWEEN ARIMA AND LSTM PRICE PREDICTION MODELS

Model	MAE	RMSE	Training Loss (MSE)	Prediction Accuracy
ARIMA	1850	2360	0.012	82.4%
LSTM	620	890	0.003	93.7%

By comparing different approaches, we found that while methods like Random Forest work well as a starting point, the LSTM structure is necessary for handling the complex and time-based changes in e-commerce prices.

V. CONCLUSION

This project led to the creation of an AI system that helps predict how online product prices might change over time and suggests the best time to buy. The system uses a type of neural network called LSTM, which is good at learning from past price data and making accurate guesses about future price changes.

By looking at how prices have changed in the past from online shopping data, the system can find trends and give useful predictions to help people make better buying choices.

The tests showed that the LSTM model made fewer mistakes compared to older methods like ARIMA when looking at the same online price data. This means that deep learning models are better at understanding the complicated patterns in how prices change over time. Besides predicting price changes, the system also gives customers clear advice, like "Buy Now" or "Wait for a Better Price," depending on what the model predicts for the future.

In general, this approach has a lot of promise for helping people make smarter choices when shopping online. Future improvements could include adding more factors like holiday trends, changes in how much people want a product, special sales, and more types of products in the data. Adding these elements could make the predictions even better and make the system work well on different online stores.

VI. DECLARATION

CONFLICT OF INTEREST The authors declare that there is no conflict of interest regarding the publication of the paper. DATA AVAILABILITY Not applicable. FUNDING Not applicable. References

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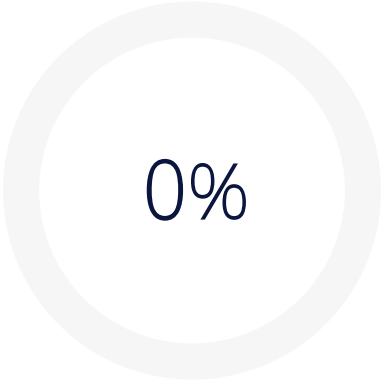

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